DECK 02 · WEEK 1 · MORNING 2

Probability, Distributions & Portfolio Math

From moments of one return series to Monte Carlo pricing of multi-asset portfolios

CFA Level I · Vol 1 Learning Modules 3–6

Roadmap — Probability, Distributions, Portfolio Mathematics & Simulation

4 sections · ~2 hours · Built bottom-up from CFA readings

01

Moments, shape & dispersion

AM / median / mode · trimmed & winsorized · variance / MAD / semideviation · skew & kurtosis · correlation

02

Probability trees & Bayes

E[X], Var[X] for discrete RVs · BankCorp EPS conditional tree · total probability rule · Bayes in credit & equity

03

Portfolio risk & safety-first

weighted return · covariance matrix · diversification · joint probability function · shortfall risk · Roy's criterion

04

Lognormal & Monte Carlo

lognormal asset prices · r_cc & √T scaling · Astra volatility · 6-step Monte Carlo · bootstrap vs MC

Learning objectives

By the end of this deck you should be able to:

1. Compute and interpret the full set of descriptive statistics for an asset-return series — mean variants, variance / MAD / semideviation, skewness, kurtosis, covariance and correlation — and know which one answers which question.

2. Build a probability tree for a discrete random variable, read off conditional expected values and variances, and apply Bayes' formula to update a prior belief with new information.

3. Decompose portfolio variance into own-variance and covariance terms, enumerate how many distinct covariances an n-asset portfolio requires, and estimate covariance from a joint probability function when history is short.

4. Quantify shortfall risk using Roy's safety-first criterion and connect the SFRatio to the Sharpe ratio as a special case when the threshold equals the risk-free rate.

5. Explain why asset prices are modelled as lognormal while continuously compounded returns are modelled as normal, derive the $\sqrt{T}$ volatility-scaling rule, and walk through the 6-step Monte Carlo recipe for a derivative with no closed form.

SECTION 01

Moments, shape & dispersion

Mean, median and mode answer three different questions about the same return series — and outliers are what force you to pick carefully

Arithmetic mean is the most used — and the least robust to outliers. Trimmed and winsorized means reclaim robustness without abandoning the mean entirely.

$$\bar{X} = (1/n)\ \Sigma\ X_i$$

median = middle value

mode = most frequent value

where: Trimmed mean: drop top & bottom k%. Winsorized mean: replace tails with boundary quantile.

EAA Equity Index, 1,258 trading days	Observations	Mean	Interpretation
Arithmetic mean (all data)	1,258	0.035%	baseline
5% trimmed (drop top / bottom 2.5%)	1,194	0.048%	higher → neg. outliers dropped
95% winsorized (cap at ±P97.5)	1,258	0.038%	modest — tails capped not dropped

WHY IT MATTERS

Gap between arithmetic, trimmed and winsorized means is a first-pass test for the presence of influential outliers. In this series the mean moves ~37% relative to itself — evidence the extreme observations are pulling the estimate.

Source: CFA Institute, Curriculum Vol 1, Learning Module 3, pp. 89–93.

Variance and standard deviation describe two-sided dispersion — downside deviation and coefficient of variation sharpen the view when asymmetry matters

Target semideviation only penalises observations below a chosen threshold — the investor-relevant cut of risk.

$$s^2 = \Sigma(X_i - \bar{X})^2 / (n-1) \qquad s_target = \sqrt{[\Sigma_{\{X_i<B\}}(X_i - B)^2 / (n-1)]}$$

$$CV = s / \bar{X}$$

where: B = target (minimum acceptable return). CV is unitless — enables direct comparison of dispersion across datasets.

CFA LM 3 EXAMPLE 3 — MONTHLY PORTFOLIO RETURNS

12 monthly returns: 5, 3, −1, −4, 4, 2, 0, 4, 3, 0, 6, 5 (%). Mean $\bar{X}$ = 2.25%.

1. Sample standard deviation: s = √(96.25 / 11) = 2.958%
2. Target = 3%: squared deviations below target sum to 84 → s_target = √(84/11) = 2.763%
3. Target = 2% (lower bar): s_target = √(36/11) = 1.809% → falls because fewer observations clear the bar
4. Coefficient of variation: CV = 2.958 / 2.25 = 1.31 → dispersion is 1.3× the mean — high relative risk

s = 2.958%; s_target@3% = 2.763%; CV = 1.31

Source: CFA Institute, Curriculum Vol 1, Learning Module 3, pp. 100–108, Ex. 3–4.

Skewness measures asymmetry and kurtosis measures tail weight — both matter because real asset returns are almost never normal

Normal = skew 0 and kurtosis 3. Excess kurtosis > 0 signals fat tails — financial returns routinely show +3 to +5 of excess kurtosis.

$$\text{Skew} \approx (1/n)\ \Sigma\ (X_i - \bar{X})^3 / s^3 \qquad \text{Excess Kurtosis} \approx (1/n)\ \Sigma\ (X_i - \bar{X})^4 / s^4 - 3$$

where: Positive skew → long right tail (rare gains, frequent small losses). Negative skew → the mirror.

NEGATIVE SKEW

Frequent small gains, rare large losses.

Typical of equity indices — crashes are the tail. Mean < median < mode.

LEPTOKURTIC · K_E > 0

More probability in tails AND near the mean; less in the shoulders.

EAA Equity Index: skew −0.43, excess kurtosis +3.80 → fat-tailed and left-skewed.

IGNORE AT YOUR PERIL

Variance-only risk models underestimate extreme losses when skew and kurtosis are non-trivial — why VaR is routinely backtested against fat-tailed empirical data.

Source: CFA Institute, Curriculum Vol 1, Learning Module 3, pp. 110–116, Ex. 6.

Covariance is scale-dependent and therefore hard to compare — correlation normalises it to the interval [−1, +1] and becomes the workhorse

Correlation of 1 = perfect positive linear. Correlation of 0 = no linear association (but could still be non-linearly related).

$$s_{XY} = \Sigma (X_i - \bar{X})(Y_i - \bar{Y}) / (n-1) \qquad r_{XY} = s_{XY} / (s_X \cdot s_Y)$$

where: Covariance is in units of X × Y; correlation is unitless and bounded by ±1.

Asset pair	Variances	Covariance	Correlation
Large-cap US vs Large-cap world (ex-US)	σ^2 0.00174 / 0.00149	0.00135	0.876
Intermediate IG corp vs Long-term IG corp	σ^2 0.00017 / 0.00070	0.00032	0.951
Equity-equity pair (lower correlation)	mixed variances	—	weaker link
Bond-bond pair (higher correlation)	lower variances	—	stronger link

Source: CFA Institute, Curriculum Vol 1, Learning Module 3, pp. 117–122.

Correlation captures only linear association — scatter plots remain essential to spot non-linearity, outliers, and spurious relationships

IT sector returns track the S&P 500 tightly; utility returns do not. The scatter plot diagnoses the relationship before the coefficient is computed.

WHAT THE PLOT REVEALS

1. Overall strength and direction of association
2. Linearity vs curvature
3. Data range and gaps
4. Outliers that may distort the coefficient
5. Clusters suggesting sub-populations or regime shifts

CORRELATION ≠ CAUSATION

Presence of a strong r_XY only implies the two variables move together historically. Causal attribution requires:

- Economic theory
- Stable regime
- Controls for confounders

Never infer causation from a scatter plot alone.

[1] A single outlier can pull a correlation coefficient from 0.1 to 0.7 — always inspect the cloud of points.

[2] Spurious correlation: two series that move with a common third factor (e.g., inflation) are not themselves causally linked.

Source: CFA Institute, Curriculum Vol 1, Learning Module 3, pp. 117–122.

SECTION 02

Probability trees & Bayes

The expected value of a discrete random variable is the probability-weighted sum of its outcomes — variance is the probability-weighted squared deviation around it

Standard deviation brings variance back into the same units as the underlying variable — usually how investors think and talk.

$$E[X] = \Sigma P(X_i) \cdot X_i \qquad \sigma^2[X] = \Sigma P(X_i) \cdot [X_i - E[X]]^2$$

$$\sigma[X] = \sqrt{\sigma^2[X]}$$

CFA LM 4 EXAMPLE 1 — BANKCORP EPS FORECAST

Four possible EPS outcomes with probabilities 0.15, 0.45, 0.24, 0.16 and EPS \$2.60 / \$2.45 / \$2.20 / \$2.00.

1. E[EPS] = 0.15(2.60) + 0.45(2.45) + 0.24(2.20) + 0.16(2.00) = \$2.3405
2. σ^2[EPS] = $0.15(2.60-2.34)^2 + 0.45(2.45-2.34)^2 + 0.24(2.20-2.34)^2 + 0.16(2.00-2.34)^2$
3. = 0.01014 + 0.00545 + 0.00470 + 0.01850 = 0.03879
4. σ[EPS] = √0.03879 ≈ \$0.197 → expected EPS of \$2.34 with one-sigma band of ±\$0.20

$$E[EPS] = \$2.34 \cdot \sigma[EPS] = \$0.20$$

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, pp. 134–136.

A probability tree decomposes an unconditional forecast into scenario-conditional expectations that must recombine to the original number

Total probability rule for expected value: $E[X] = \Sigma\, E[X \mid S_i] \cdot P(S_i)$. The check that the recombined total matches the unconditional $E[X]$ is non-negotiable.

$$E[X] = E[X \mid S_1] \cdot P(S_1) + E[X \mid S_2] \cdot P(S_2) + \ldots + E[X \mid S_n] \cdot P(S_n)$$

where: Scenarios $S_1..S_n$ must be mutually exclusive and exhaustive. Their probabilities sum to 1.

Scenario	$P(S_i)$	$E[EPS \mid S_i]$	Contribution	Check
Declining interest rates	0.60	0.25·2.60+0.75·2.45 = $2.4875	1.4925	✓
Stable interest rates	0.40	0.60·2.20+0.40·2.00 = $2.1200	0.8480	✓
Unconditional E[EPS]	**1.00**	**—**	**$2.3405**	**= Slide 11**

CONDITIONAL EXPECTED VALUE IS HOW ANALYSTS COMMUNICATE

"Under a recession, we expect EPS of $2.12; under our base case, $2.49." — the scenario-conditioned forecast tells a richer story than a single point estimate.

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, pp. 136–140.

Conditional variance is how scenario-conditioned forecasts keep track of residual risk inside each branch of the tree

Unconditional variance = average of conditional variances + variance of conditional means (law of total variance, discussed later in curriculum).

$$\sigma^2[X \mid S] = \Sigma_i P(X_i \mid S) \cdot [X_i - E[X \mid S]]^2$$

where: Computed exactly like unconditional variance, but using probabilities conditional on scenario S and deviations from the conditional mean E[X | S].

DECLINING RATES SCENARIO

Conditional E[EPS] = $2.49

$\sigma^2 = 0.25(2.60-2.49)^2 + 0.75(2.45-2.49)^2$
$= 0.00422$

$\sigma = \$0.065$

STABLE RATES SCENARIO

Conditional E[EPS] = $2.12

$\sigma^2 = 0.60(2.20-2.12)^2 + 0.40(2.00-2.12)^2$
$= 0.00960$

$\sigma = \$0.098$ → about 50% more volatile than the declining-rates branch

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, pp. 138–140.

Bayes' formula reverses a conditional probability — it updates a prior belief by the ratio of likelihood to unconditional probability of the new information

The multiplier [P(Info | Event) / P(Info)] is the "evidence ratio". When it is >1 the posterior rises; <1 and the posterior falls.

$$P(\text{Event} \mid \text{Info}) = [\, P(\text{Info} \mid \text{Event}) / P(\text{Info}) \,] \cdot P(\text{Event})$$

where: P(Event) = prior; P(Info | Event) = likelihood; P(Info) = unconditional probability; P(Event | Info) = posterior.

TECH-FIRM FRAMEWORK — READ THE UPDATE AS A FREQUENCY

A 500-stock index has 100 tech firms; 60 had returns >10%. Of 400 non-tech firms, 100 had returns >10%. What is P(tech | R > 10%)?

1. Prior: P(tech) = 100/500 = 0.20 · unconditional P(R>10%) = 160/500 = 0.32
2. Likelihood: P(R>10% | tech) = 60/100 = 0.60
3. Bayes: P(tech | R>10%) = (0.60 / 0.32) · 0.20 = 1.875 · 0.20 = 0.375
4. Frequency check: of 160 firms with R>10%, 60 are tech → 60/160 = 0.375 ✓

Posterior 0.375 ≈ prior 0.20 × evidence ratio 1.875

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, pp. 141–144.

The DriveMed expansion announcement lifts the implied probability of a consensus beat from 45% to 82% — a textbook Bayesian update

Three mutually exclusive events (beat / meet / miss) — the three posteriors must sum to 1, which is the final consistency check on the calculation.

Event	Prior	P(expansion | Event)	Joint	Posterior
EPS exceeded consensus	0.45	0.75	0.3375	0.823
EPS met consensus	0.30	0.20	0.0600	0.146
EPS fell short	0.25	0.05	0.0125	0.030
P(expansion) · Σ posterior	**1.00**	**–**	**0.4100**	**1.000**

MECHANICS

Posterior = Joint / P(expansion) — for example P(beat | expands) = 0.3375 / 0.41 = 0.823. A single piece of information (the factory expansion) shifts the probability mass dramatically from 45 / 30 / 25 to 82 / 15 / 3.

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, Example 4, pp. 144–147.

Bayes is most valuable where the base rate is low and the signal is imperfect — credit, bankruptcy prediction, manager skill evaluation

These three practice problems share the same structure — priors, likelihoods, posterior via Bayes — and together anchor the formula in realistic analyst workflows.

CREDIT SCORE · TIMELY REPAYMENT

Priors:
P(A) = 0.90 (pays on time)
P(B) = 0.80 (good credit)

Likelihood:
P(B | A) = 0.85

Posterior:
P(A | B) = 0.85 / 0.80 × 0.90
= 0.956

Good credit report raises timely-pay probability from 90% to 95.6%.

BANKRUPTCY MODEL · FAILURE RATE

Priors:
P(fail) = 0.20
P(good) = 0.70

Likelihood:
P(good | fail) = 0.50

Posterior:
P(fail | good) = 0.50 / 0.70 × 0.20
= 0.143

A "good" rating drops the failure probability from 20% to 14.3%.

CEO FIRED · GOOD PERFORMANCE

Priors:
P(fired) = 0.05
P(good) = 0.50

Likelihood:
P(good | fired) = 0.30

Posterior:
P(fired | good) = 0.30 / 0.50 × 0.05
= 0.030

Even good performance leaves a 3% chance of firing — boards fire for reasons other than stock return.

Source: CFA Institute, Curriculum Vol 1, Learning Module 4, Question Set, pp. 148–151.

SECTION 03

Portfolio risk

A portfolio's expected return is a simple weighted average, but its variance depends on pairwise covariances — which quickly dominate as n grows

For n assets there are n variance terms and n(n−1)/2 distinct covariance terms — at n = 20 the ratio is 1:19 in favour of covariances.

$$E[R_p] = \Sigma\ w_i\ E[R_i] \qquad \sigma^2[R_p] = \Sigma_i\ \Sigma_j\ w_i\ w_j\ Cov(R_i, R_j)$$

where: Weights sum to 1. Diagonal $Cov(R_i, R_i) = \sigma^2[R_i]$. Off-diagonal terms capture co-movement.

CFA LM 5 — THREE-ASSET PORTFOLIO (S&P 500 / LT CORP BONDS / MSCI EAFE)

Weights 50% / 25% / 25%. E[R] = 13% / 6% / 15%. Var diagonals 400 / 81 / 441 (%²). Off-diagonals 45 / 189 / 38.

1. E[R_p] = 0.50(13) + 0.25(6) + 0.25(15) = 11.75%
2. Variance contribution (diagonals): 0.25·400 + 0.0625·81 + 0.0625·441 = 132.625 (%²)
3. Covariance contribution: 2·0.50·0.25·45 + 2·0.50·0.25·189 + 2·0.25·0.25·38 = 63.25 (%²)
4. σ^2[R_p] = 195.875 → σ[R_p] = 14.00% (vs 11.52% if covariances were zero — 2.5 pp of risk from co-movement)

E[R_p] = 11.75% · σ[R_p] = 14.00% (diversification gap ~250 bp)

Source: CFA Institute, Curriculum Vol 1, Learning Module 5, pp. 154–157.

The covariance matrix captures everything portfolio math needs — for n assets it contains n variances plus n(n−1)/2 distinct covariances

The covariance matrix is symmetric (Cov(A,B) = Cov(B,A)), so only the upper (or lower) triangle needs to be estimated.

Inputs	Variances	Distinct covariances	Total
n = 3	3	3	6
n = 5	5	10	15
n = 10	10	45	55
n = 20	20	190	210
n = 50	50	1,225	1,275
n = 100	100	4,950	5,050

PRACTICAL IMPLICATIONS

Estimation becomes the bottleneck long before the formulas do. For a 500-stock portfolio you need 124,750 covariance estimates — far more than return history usually supports.

Consequence: factor models (market + style factors) and shrinkage estimators exist precisely to make the covariance matrix estimable and well-conditioned.

Source: CFA Institute, Curriculum Vol 1, Learning Module 5, pp. 156–160.

A two-asset portfolio traces the efficient frontier — low correlation does the diversification work, not low variance

The minimum-variance combination for ρ = 0.30 is ~85% Stock 1 / 15% Stock 2 — less than either standalone standard deviation.

$$\sigma^2[R_p] = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2 \cdot \rho \cdot \sigma_1 \cdot \sigma_2$$

where: With ρ < 1, σ[R_p] is strictly below the weighted average of σ_1 and σ_2 — this is the diversification benefit.

w_1 / w_2	E[R_p] (%)	σ^2[R_p] (%²)	σ[R_p] (%)	vs linear σ
1.00 / 0.00	4.00	36.00	6.00	—
0.80 / 0.20	4.80	40.68	6.38	−1.42 pp
0.60 / 0.40	5.60	61.92	7.87	−1.73 pp
0.40 / 0.60	6.40	99.72	9.99	−1.61 pp
0.20 / 0.80	7.20	154.08	12.41	−0.79 pp
0.00 / 1.00	8.00	225.00	15.00	—

Source: CFA Institute, Curriculum Vol 1, Learning Module 5, Example 1, pp. 159–161.

A joint probability function converts discrete scenarios into a forward-looking covariance — the right tool when history is short or regime-shifting

If X and Y are independent then P(X,Y) = P(X)·P(Y) and Cov(X,Y) = 0 — but the converse (zero correlation ⇏ independence) does NOT hold.

$$\mathrm{Cov}(R_A, R_B) = \Sigma_i \Sigma_j \; P(R_{A,i}, R_{B,j}) \cdot (R_{A,i} - E[R_A]) \cdot (R_{B,j} - E[R_B])$$

where: Sum over all discrete (i, j) outcomes weighted by their joint probability — the discrete analogue of the continuous covariance integral.

CFA LM 5 — BANKCORP × NEWBANK JOINT DISTRIBUTION

Three banking-industry states: good (0.20), average (0.50), poor (0.30). BankCorp returns 25 / 12 / 10%; NewBank returns 20 / 16 / 10%.

1. E[R_A] = 0.20(25) + 0.50(12) + 0.30(10) = 14% · E[R_B] = 0.20(20) + 0.50(16) + 0.30(10) = 15%
2. Good: (25−14)(20−15) = 55 → weighted 0.20 · 55 = 11
3. Average: (12−14)(16−15) = −2 → weighted 0.50 · (−2) = −1
4. Poor: (10−14)(10−15) = 20 → weighted 0.30 · 20 = 6

$$\mathrm{Cov}(R_A, R_B) = 11 - 1 + 6 = 16 \quad (\%^2)$$

Source: CFA Institute, Curriculum Vol 1, Learning Module 5, pp. 163–166.

Shortfall risk — the probability of falling below a minimum acceptable return — is the downside metric that matches investor objectives

Symmetric σ treats upside and downside equally. Shortfall risk asks the only question clients care about: what is the chance I lose more than I can afford?

WHY SYMMETRIC RISK MISSES

Two portfolios with identical σ but different mean returns have very different shortfall probabilities under normality.

With target R_L, the portfolio with the higher mean has a smaller shaded area to the left of R_L — the same σ, but a dramatically different lived experience.

THREE PORTFOLIOS AT R_L = 2%

Portfolio I: E = 5%, σ = 8% → P(x<2%) = 37.7%
Portfolio II: E = 8%, σ = 8% → P(x<2%) = 24.6%
Portfolio III: E = 5%, σ = 12% → P(x<2%) = 41.7%

II dominates I (higher return, same σ) and III (same return, lower σ). Both effects matter.

Source: CFA Institute, Curriculum Vol 1, Learning Module 5, pp. 167–169.

Roy's safety-first criterion picks the portfolio that maximises the distance from the mean to the threshold, expressed in standard deviations

Under normal returns, $P(R_p < R_L) = N(-SFRatio)$ — so maximising the SFRatio minimises the shortfall probability. When $R_L = R_f$, SFRatio = Sharpe.

$$SFRatio = (E[R_p] - R_L) / \sigma[R_p]$$ pick the portfolio with the highest SFRatio

where: R_L = minimum acceptable return. For normal returns, shortfall probability = $N(-SFRatio)$.

CFA LM 5 EXAMPLE 3 — CAD 800,000 CLIENT · WITHDRAW CAD 30,000 IN 1Y

R_L = 30,000 / 800,000 = 3.75%. Three allocations A / B / C with E / σ = 25% / 27%, 11% / 8%, 14% / 20%.

1. A: SFRatio = (25 − 3.75) / 27 = 0.787
2. B: SFRatio = (11 − 3.75) / 8 = 0.906 ← highest
3. C: SFRatio = (14 − 3.75) / 20 = 0.513
4. Shortfall probability for B under normality: N(−0.906) ≈ 18.2%

Allocation B wins on Roy's criterion — ~18% chance of invading principal despite only 11% expected return

Source: CFA Institute, Curriculum Vol 1, Learning Module 5, Example 3, pp. 169–170.

SECTION 04

Lognormal & Monte Carlo

Asset prices are modelled as lognormal because they are bounded below by zero and right-skewed — properties the normal distribution cannot reproduce

If ln Y is normal, then Y is lognormal. The lognormal mean exceeds exp(μ) by a convexity term that grows with σ.

$$E[Y] = \exp(\mu + 0.5\,\sigma^2) \qquad Var[Y] = \exp(2\mu + \sigma^2) \cdot [\exp(\sigma^2) - 1]$$

where: μ, σ^2 are the mean and variance of the associated normal distribution (ln Y). Used by Black–Scholes–Merton.

BOUNDED BELOW BY ZERO

Stock prices cannot go negative. Normal distribution assigns probability to negative prices; lognormal does not.

RIGHT-SKEWED

Long right tail reflects compounding upside — a single year of +100% can double wealth, whereas −100% is the loss floor.

CONVEXITY PREMIUM

$E[\exp(X)] > \exp(E[X])$ by factor $\exp(0.5\sigma^2)$. As volatility rises, the mean of prices drifts further above the median. Jensen's inequality in action.

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 178–181.

Continuously compounded returns are additive across time, so their sum over T periods scales linearly in mean and variance — the √T rule for volatility

Independence + identical distribution (iid) is the assumption. When it breaks (autocorrelation, regime shifts), √T scaling breaks too.

$$r_{0,T} = \Sigma\, r_t \qquad E[r_{0,T}] = \mu \cdot T \qquad \sigma^2[r_{0,T}] = \sigma^2 \cdot T$$

$$\sigma[r_{0,T}] = \sigma \cdot \sqrt{T}$$

where: Under iid continuously compounded per-period returns with mean μ and variance σ².

Period (under σ_daily = 1%)	T (days)	σ_T	Annualised σ
1 trading day	1	1.00%	15.81%
1 week (5 trading days)	5	2.24%	15.81%
1 month (~21 trading days)	21	4.58%	15.81%
1 quarter (~63 trading days)	63	7.94%	15.81%
1 year (250 trading days)	250	15.81%	15.81%

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 181–183.

A week of Astra International closing prices lets us compute daily continuously compounded returns — √250 scaling gives an annualised volatility of 33.6%

4 return observations — far too few to estimate expected return but enough to illustrate the mechanical volatility-annualization pipeline.

Day	Close (IDR)	$\ln(P_t/P_{t-1})$
Mon	6,950	–
Tue	7,000	+0.00717
Wed	6,850	−0.02166
Thu	6,600	−0.03718
Fri	6,350	−0.03862

ANNUALIZATION PIPELINE

4 daily returns → sample σ → × √250 to annualize.

1. Mean $\bar{r}$ = −0.02257 (ignore — sample too short)
2. Sample variance s^2 = 0.000452 (divisor n−1 = 3)
3. Daily σ = √0.000452 = 0.02126 = 2.13%
4. Annualized σ = 0.02126 × √250 = 0.3362 = 33.6%

Annualized volatility ≈ 33.6%

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, Example 1, pp. 182–184.

Monte Carlo simulation values securities where no closed form exists by generating thousands of random price paths from assumed distributions

Specify once (steps 1–3), run repeatedly (steps 4–6). The more trials, the tighter the statistical estimate.

1 SPECIFY QUANTITY OF INTEREST
What are we valuing? E.g., C_{iT} = value of an Asian option at maturity given simulated stock-price path i.

2 SPECIFY TIME GRID
Divide the horizon into K sub-periods of length Δt. Example: 1 year with K = 12 gives monthly steps.

3 SPECIFY DATA-GENERATION METHOD
Distributional assumption for each risk factor (e.g., standard normal Z_k for stock shocks); pick μ, σ.

4 CONVERT RANDOM DRAWS TO PRICES
Apply the price-process model, e.g. $\Delta S = \mu S \Delta t + \sigma S Z_k$, iterated across the K sub-periods to build one path.

5 COMPUTE PAYOFF FOR ONE TRIAL
Calculate the claim's terminal payoff C_{iT}; discount to C_{i0} using an appropriate rate. One complete trial is done.

6 REPEAT N TRIALS, SUMMARISE
Typically N = 1,000 to 100,000 trials. Average of C_{i0} across trials is the Monte Carlo estimate of the claim's present value.

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 184–187.

Valuing an Asian option illustrates Monte Carlo end-to-end — average-minus-final payoff, 12 monthly steps, 1,000 trials

Payoff = max(S_T − average(S), 0). No closed form exists because the payoff depends on the path, not only the terminal stock price.

$$\Delta S = \mu \cdot S \cdot \Delta t + \sigma \cdot S \cdot Z_k \qquad C_{iT} = \max(S_T - \bar{S}, 0)$$

$$\text{Value} \approx (1/N)\ \Sigma_i\ e^{(-rT)} \cdot C_{iT}$$

where: Z_k ~ standard normal i.i.d. $\bar{S}$ = arithmetic average of the K simulated prices.

ILLUSTRATIVE RUN — 1,000 TRIALS, T = 1Y, K = 12

Draw 12,000 standard-normal shocks; build 1,000 paths of 12 monthly prices each; compute payoff, discount, average.

1. Path A: S_T = USD 34, $\bar{S}$ = USD 31 → payoff max(3, 0) = USD 3
2. Path B: S_T = USD 28, $\bar{S}$ = USD 32 → payoff max(−4, 0) = USD 0
3. Across 1,000 paths: 346 pay strictly positive (34.6%), 654 pay zero
4. Discount each C_iT at the risk-free rate, take mean → Monte Carlo estimate of the option value today

Max simulated payoff = USD 11 · option PV estimate = average of discounted C_iT

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, Ex. 4–5 and Example 2, pp. 185–189.

Bootstrap resampling swaps the assumed distribution for the historical one — same recipe, but draws with replacement from the observed sample

Bootstrap answers "what if history repeats?" Monte Carlo answers "what if my distributional assumption is correct?" Both are complements to closed-form pricing.

Dimension	Monte Carlo	Bootstrap
Source of random draws	Parametric distribution (e.g., normal)	Observed historical sample (with replacement)
Distributional assumption	Required — must pick μ, σ, family	None — empirical distribution is taken as truth
Strength	Handles any custom payoff or model structure	Captures real-world fat tails and asymmetries for free
Weakness	Wrong assumption → wrong answer	Only as good as the sample — cannot extrapolate beyond observed
When to use	Pricing derivatives, stress testing, scenarios	Resampling for confidence intervals, back-testing, non-parametric inference

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 189–192.

Monte Carlo and bootstrap give only statistical estimates — analytical solutions remain preferable whenever they exist because they show cause-and-effect

Variance of the Monte Carlo estimate shrinks as 1/N — so a 1 percentage-point tightening of the confidence band requires roughly 100× the trials.

USE ANALYTICAL

- European options → Black–Scholes
- Vanilla bonds → DCF
- Normally-distributed portfolio VaR → parametric

Why:
— Exact
— Cheap
— Reveals cause-and-effect

USE MONTE CARLO

- Path-dependent payoffs (Asian, barrier, lookback)
- MBS with prepayment models
- Portfolio VaR with non-normal risk factors

Why:
— Flexible
— Handles any model
— Quantifies tail risk

USE BOOTSTRAP

- Confidence intervals on a complicated estimator
- Back-testing strategies over history
- Stress tests anchored in realised data

Why:
— No distribution assumed
— Fat tails preserved
— Simple to implement

[1] Statistical estimates carry sampling error that decreases as √N — doubling accuracy requires 4× the trials.
[2] Analytical solutions provide insight into how parameters drive the answer (comparative statics) — simulations do not.

Source: CFA Institute, Curriculum Vol 1, Learning Module 6, pp. 186–193.

KEY TAKEAWAYS

Deck 2 — Probability, Distributions, Portfolio Mathematics & Simulation

- Central tendency, dispersion, skewness, kurtosis and correlation are the five descriptive lenses every return series needs — each answers a different question.
- Probability trees + total probability rule decompose unconditional expectations into scenario-conditional ones; variance has a conditional counterpart too.
- Portfolio variance = own variances + n(n−1)/2 covariances; diversification shows up when $\rho < 1$. A 20-asset portfolio has 190 distinct covariance inputs.
- Prices lognormal, continuously compounded returns normal; volatility scales with $\sqrt{T}$ under iid; Astra IDR weekly example → 33.6% annualised σ.
- Target semideviation and coefficient of variation refine the σ view; skew and excess kurtosis expose the non-normality that every equity series exhibits.
- Bayes' formula updates priors via the evidence ratio [P(Info|Event) / P(Info)]; DriveMed expansion raised P(EPS beat) from 45% to 82%.
- Roy's safety-first criterion: maximise SFRatio = (E[R_p] − R_L)/σ[R_p]. When R_L = R_f it becomes the Sharpe ratio — same formula, different framing.
- Monte Carlo values path-dependent securities (Asian / lookback / MBS) by averaging discounted payoffs over 1,000+ simulated paths; bootstrap is its non-parametric cousin.